

Tailored Application Package: AI Research & Systems Intern

Applicant: Alex Rivera

Email: alex.rivera@cs.stanford.edu | **GitHub:** github.com/arivera-ai

Target Role: AI Research & Systems Intern (AnthroMetrics AI Research)

Date: September 2026

1. Executive Skill Gap & Alignment Analysis

Competency Area	AnthroMetrics JD Requirement	Alex Rivera Experience & Match	Alignment Level
Local LLM & Inference	Experience with quantized SLMS, vLLM, Ollama, ONNX	Engineered sub-80ms semantic RAG engine at VeriLocal Systems; benchmarked 7B SLMS at Cognition Labs	EXACT MATCH (100%)
Agent Safety & Verification	Sandboxed tool-use, safety rails, verifiable execution	Created SecureAgent runtime; built human permission gates & filesystem post-action verifiers	EXACT MATCH (100%)
Retrieval & RAG	Hybrid sparse-dense retrieval, ChromaDB / FAISS	Implemented hybrid RRF search in Project Alpha; indexed 50k+ internal documents	EXACT MATCH (100%)
Systems & Systems Code	Python, PyTorch, C++/Rust, AsyncIO	Advanced Python, intermediate Rust, B.S. CS from Stanford University (GPA: 3.92)	HIGH MATCH (95%)

Key Strategic Highlights

- **Direct Systems Background:** Built deterministic sandboxing frameworks reducing unauthorized file access to 0%.
- **Relevant Stanford Coursework:** CS224N (Natural Language Processing), CS231N (Computer Vision), CS229 (Machine Learning), CS110 (Computer Systems).
- **Quantized Benchmark Expertise:** Prior experience evaluating tool precision and JSON adherence in 7B/8B local models.

2. Customized Cover Letter

To: Hiring Team, AnthroMetrics AI Research

Subject: Application for AI Research & Systems Intern — Alex Rivera

Dear Hiring Team at AnthroMetrics AI,

I am writing to express my enthusiastic interest in the AI Research & Systems Intern position. Having tracked AnthroMetrics' pioneering contributions to verifiable intelligence and reliable agent architectures, I believe my background in local inference optimization, deterministic tool sandboxing, and hybrid retrieval systems aligns directly with your

mission.

During my time as an AI Systems Engineer at VeriLocal Systems, I spearheaded the development of an on-device RAG engine indexing over 50,000 documents with sub-80ms latency. More importantly, I designed and implemented human-in-the-loop permission gateways and post-action disk verifiers, guaranteeing that autonomous actions remain fully deterministic, inspectable, and secure.

Prior to that, during my internship at Cognition Labs, I developed quantitative evaluation frameworks for local quantized models (Qwen, Llama, Mistral), achieving 94.2% valid tool-dispatch syntax. These practical engineering experiences, coupled with my Stanford Computer Science foundation (GPA: 3.92), have given me a rigorous understanding of both the theoretical and systems-level challenges inherent to trustworthy agent design.

I would welcome the opportunity to contribute to AnthroMetrics' state-of-the-art research and systems initiatives. Thank you for your time and consideration.

Sincerely,

Alex Rivera

alex.rivera@cs.stanford.edu | github.com/arivera-ai

3. 4-Week Project & Onboarding Plan

- **Week 1: Architecture Onboarding & Benchmark Baselines**
 - Profile existing model inference pipelines using Ollama/vLLM across target hardware.
 - Establish automated test harnesses for structured tool call dispatch.
 - **Week 2: Verification Engine & Sandbox Integration**
 - Implement cryptographic post-action verification hooks for workspace mutations.
 - Deploy interactive human approval gates with diff-preview capabilities.
 - **Week 3: Hybrid Retrieval & RAG Optimization**
 - Integrate Reciprocal Rank Fusion (RRF) sparse-dense retrieval over internal documentation.
 - Benchmark retrieval precision and memory footprint against baseline FAISS/ChromaDB indices.
 - **Week 4: Synthesis & Production Verification**
 - Perform stress tests on concurrent agent execution and audit log immutability.
 - Package deliverables with full unit test coverage and reproducible documentation.
-

Generated and verified on-device by LocalGPT Deterministic Workspace Agent.